WEB PROGRAMMING

Laboratory Experiment 4

Recitation Document

Code + Output Screenshots

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Question 1

■ Source Code

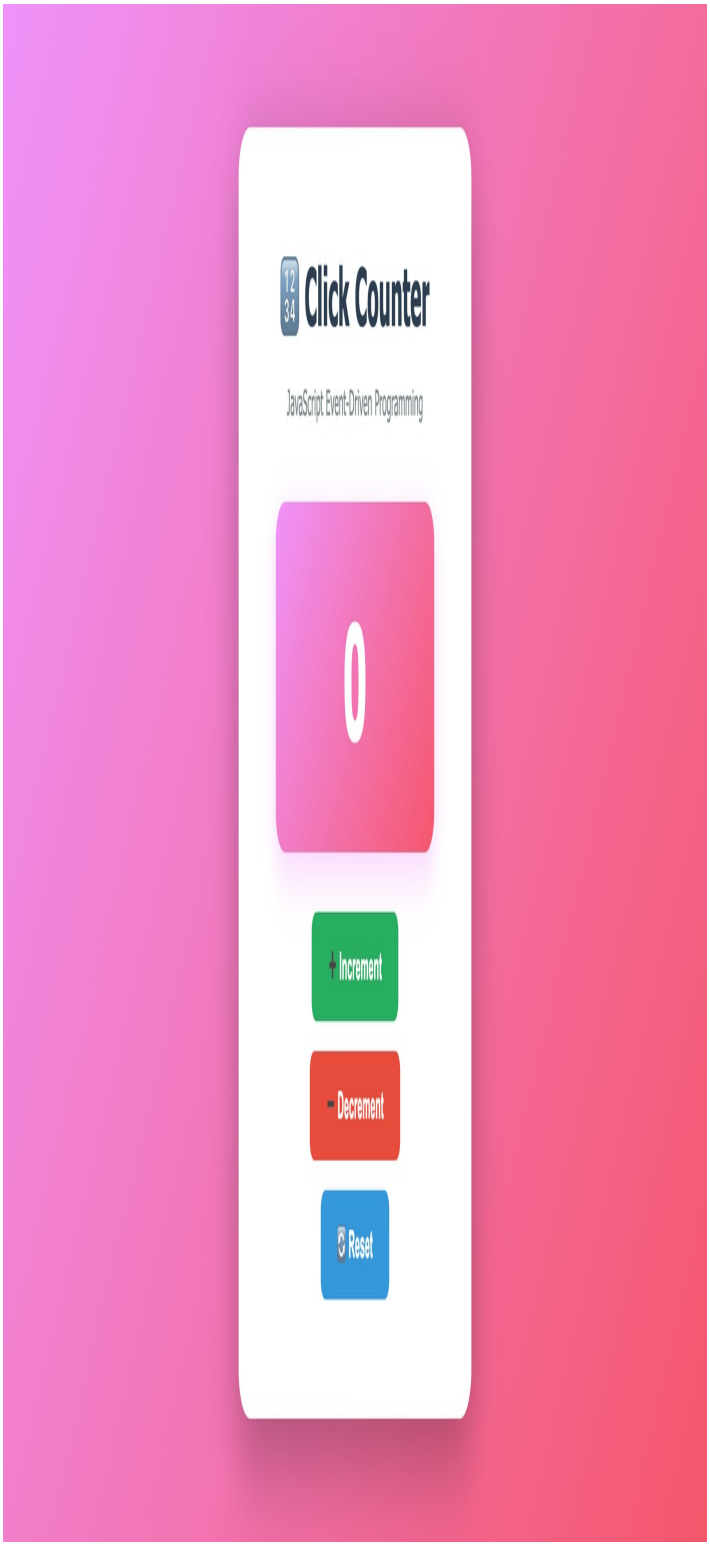
```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>Click Counter - JavaScript Events</title> <style>
* { margin: 0; padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma, Geneva,
Verdana, sans-serif; background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%); min-height: 100vh;
display: flex; align-items: center; justify-content: center; padding: 20px; } .container { background:
white; padding: 60px 80px; border-radius: 25px; box-shadow: 0 20px 60px rgba(0,0,0,0.3); text-align:
center; max-width: 500px; width: 100%; } h1 { font-size: 2.5em; color: #2c3e50; margin-bottom: 15px; }
.subtitle { color: #7f8c8d; font-size: 1.1em; margin-bottom: 40px; } .counter-display { background:
linear-gradient(135deg, #f093fb 0%, #f5576c 100%); color: white; font-size: 5em; font-weight: bold;
```

```
padding: 40px; border-radius: 20px; margin: 30px 0; box-shadow: 0 10px 30px rgba(240, 147, 251, 0.4);
transition: all 0.3s ease; } .counter-display.animate { transform: scale(1.1); } .button-group { display:
flex; gap: 15px; justify-content: center; flex-wrap: wrap; } button { background: #2c3e50; color: white;
border: none; padding: 15px 35px; font-size: 1.2em; border-radius: 10px; cursor: pointer; transition: all
0.3s ease; font-weight: 600; } button:hover { background: #34495e; transform: translateY(-3px);
box-shadow: 0 8px 20px rgba(0,0,0,0.2); } button:active { transform: translateY(0); } .increment {
background: #27ae60; } .increment:hover { background: #229954; } .decrement { background: #e74c3c; }
.decrement:hover { background: #c0392b;
```

```
} .reset { background: #3498db; } .reset:hover { background: #2980b9; } @media (max-width: 600px) {
.container { padding: 40px 30px; } .counter-display { font-size: 3.5em; padding: 30px; } button { padding:
12px 25px; font-size: 1em; } } </style> </head> <body> <div class="container"> <h1>■ Click Counter</h1> <p
class="subtitle">JavaScript Event-Driven Programming</p> <div class="counter-display" id="counter">0</div>
<div class="button-group"> <button class="increment" onclick="increment()">■ Increment</button> <button
class="decrement" onclick="decrement()">■ Decrement</button> <button class="reset"
onclick="resetCounter()">■ Reset</button> </div> </div> <script> let count = 0; function animateCounter()
{ const display = document.getElementById('counter'); display.classList.add('animate'); setTimeout(() =>
display.classList.remove('animate'), 300); }
```

```
function increment() { count++; document.getElementById('counter').textContent = count; animateCounter();
} function decrement() { count--; document.getElementById('counter').textContent = count;
animateCounter(); } function resetCounter() { count = 0; document.getElementById('counter').textContent =
count; animateCounter(); } </script> </body> </html>
```

■■ Output Screenshot


Click Counter
JavaScript Event-Driven Programming
0
+ Increment
- Decrement
Reset

Question 2

■ Source Code

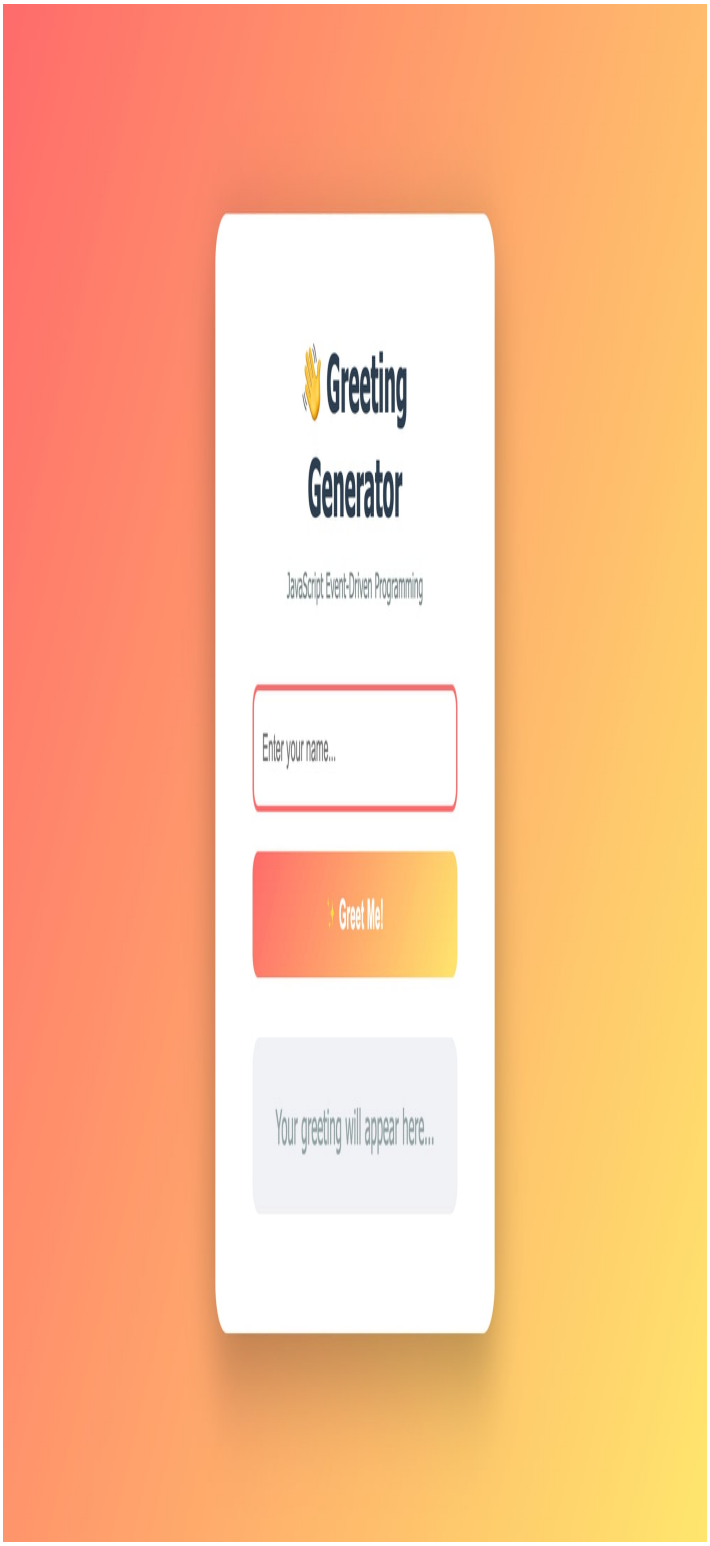
```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>Greeting Generator - JavaScript Events</title>
<style> * { margin: 0; padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma,
Geneva, Verdana, sans-serif; background: linear-gradient(135deg, #FF6B6B 0%, #FFE66D 100%); min-height:
100vh; display: flex; align-items: center; justify-content: center; padding: 20px; } .container {
background: white; padding: 60px 80px; border-radius: 25px; box-shadow: 0 20px 60px rgba(0,0,0,0.3);
text-align: center; max-width: 600px; width: 100%; } h1 { font-size: 2.5em; color: #2c3e50; margin-bottom:
15px; } .subtitle { color: #7f8c8d; font-size: 1.1em; margin-bottom: 40px; } input { width: 100%; padding:
18px; font-size: 1.2em; border: 3px solid #FF6B6B;
```

```
border-radius: 12px; margin-bottom: 20px; transition: all 0.3s ease; } input:focus { outline: none;
border-color: #FFE66D; box-shadow: 0 0 20px rgba(255, 107, 107, 0.3); } button { background:
linear-gradient(135deg, #FF6B6B 0%, #FFE66D 100%); color: white; border: none; padding: 18px 50px;
font-size: 1.3em; border-radius: 12px; cursor: pointer; transition: all 0.3s ease; font-weight: 600;
width: 100%; } button:hover { transform: translateY(-3px); box-shadow: 0 10px 30px rgba(255, 107, 107,
0.4); } button:active { transform: translateY(0); } #greetOutput { margin-top: 30px; padding: 30px;
background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); color: white; border-radius: 15px;
font-size: 1.5em; min-height: 80px; display: flex; align-items: center; justify-content: center;
box-shadow: 0 10px 30px rgba(102, 126, 234, 0.3); transition: all 0.3s ease; } #greetOutput:empty {
background: #f0f2f5; box-shadow: none;
```

```
} #greetOutput:empty:before { content: 'Your greeting will appear here...'; color: #95a5a6; font-size:
1em; } .emoji { font-size: 2em; margin-right: 15px; } @media (max-width: 600px) { .container { padding:
40px 30px; } h1 { font-size: 2em; } button, input { font-size: 1em; padding: 15px; } } </style> </head>
<body> <div class="container"> <h1>■ Greeting Generator</h1> <p class="subtitle">JavaScript Event-Driven
Programming</p> <input type="text" id="nameInput" placeholder="Enter your name..." autocomplete="off">
<button onclick="greet()">■ Greet Me!</button> <div id="greetOutput"></div> </div> <script> function
greet() { const name = document.getElementById('nameInput').value.trim(); const hour = new
Date().getHours(); let greeting, emoji; if (hour < 12) { greeting = 'Good Morning'; emoji = '■'; } else if
(hour < 17) {
```

```
greeting = 'Good Afternoon'; emoji = '■■'; } else { greeting = 'Good Evening'; emoji = '■'; } const output
= document.getElementById('greetOutput'); if (name) { output.innerHTML = `<span
class="emoji">${emoji}</span><div><strong>${greeting}, ${name}!</strong><br>Welcome to my
portfolio.</div>`; } else { output.innerHTML = `<div>Please enter your name! ■</div>`; } } // Allow Enter
key to submit document.getElementById('nameInput').addEventListener('keypress', function(event) { if
(event.key === 'Enter') { greet(); } }); </script> </body> </html>
```

■■ Output Screenshot


Greeting
Generator
JavaScript Event-Driven Programming
Enter your name...
Greet Me!
Your greeting will appear here...

Question 3

■ Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>Simple Calculator - JavaScript Events</title>
<style> * { margin: 0; padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma,
Geneva, Verdana, sans-serif; background: linear-gradient(135deg, #11998e 0%, #38ef7d 100%); min-height:
100vh; display: flex; align-items: center; justify-content: center; padding: 20px; } .container {
background: white; padding: 60px 80px; border-radius: 25px; box-shadow: 0 20px 60px rgba(0,0,0,0.3);
text-align: center; max-width: 550px; width: 100%; } h1 { font-size: 2.5em; color: #2c3e50; margin-bottom:
15px; } .subtitle { color: #7f8c8d; font-size: 1.1em; margin-bottom: 40px; } .calc-row { display: flex;
gap: 15px; margin-bottom: 25px; align-items: center;
```

```
justify-content: center; } input[type="number"] { width: 150px; padding: 18px; font-size: 1.3em; border:
3px solid #11998e; border-radius: 12px; text-align: center; transition: all 0.3s ease; }
input[type="number"]:focus { outline: none; border-color: #38ef7d; box-shadow: 0 0 20px rgba(17, 153, 142,
0.3); } select { padding: 18px 15px; font-size: 1.5em; border: 3px solid #11998e; border-radius: 12px;
background: white; cursor: pointer; transition: all 0.3s ease; } select:focus { outline: none;
border-color: #38ef7d; } button { background: linear-gradient(135deg, #11998e 0%, #38ef7d 100%); color:
white; border: none; padding: 18px 50px; font-size: 1.3em; border-radius: 12px; cursor: pointer;
transition: all 0.3s ease; font-weight: 600; width: 100%; margin-top: 10px; } button:hover { transform:
translateY(-3px); box-shadow: 0 10px 30px rgba(17, 153, 142, 0.4);
```

```
} button:active { transform: translateY(0); } #result { margin-top: 30px; padding: 30px; background:
linear-gradient(135deg, #667eea 0%, #764ba2 100%); color: white; border-radius: 15px; font-size: 2.5em;
font-weight: bold; min-height: 100px; display: flex; align-items: center; justify-content: center;
box-shadow: 0 10px 30px rgba(102, 126, 234, 0.3); transition: all 0.3s ease; } #result:empty { background:
#f0f2f5; box-shadow: none; } #result:empty:before { content: '= ?'; color: #95a5a6; font-size: 0.6em; }
@media (max-width: 600px) { .container { padding: 40px 30px; } .calc-row { flex-direction: column; }
input[type="number"] { width: 100%; } } </style> </head> <body> <div class="container"> <h1>■ Simple
Calculator</h1> <p class="subtitle">JavaScript Event-Driven Programming</p>
```

```
<div class="calc-row"> <input type="number" id="num1" placeholder="Number 1" value=""> <select
id="operator"> <option value="+">+</option> <option value="-">−</option> <option value="*">×</option>
<option value="/">÷</option> </select> <input type="number" id="num2" placeholder="Number 2" value="">
</div> <button onclick="calculate()">■ Calculate</button> <div id="result"></div> </div> <script> function
calculate() { const num1 = parseFloat(document.getElementById('num1').value); const num2 =
parseFloat(document.getElementById('num2').value); const operator =
document.getElementById('operator').value; const resultDiv = document.getElementById('result'); if
(isNaN(num1) || isNaN(num2)) { resultDiv.textContent = 'Please enter valid numbers!';
resultDiv.style.background = 'linear-gradient(135deg, #e74c3c 0%, #c0392b 100%)'; return; } let result;
switch(operator) { case '+': result = num1 + num2; break; case '-': result = num1 - num2; break; case '*':
result = num1 * num2; break; case '/': if (num2 === 0) { resultDiv.textContent = 'Cannot divide by zero!';
resultDiv.style.background = 'linear-gradient(135deg, #e74c3c 0%, #c0392b 100%)'; return; } result = num1
/ num2; break; }
```

```
result = Math.round(result * 100000) / 100000; // Round to 5 decimals resultDiv.textContent = `=
${result}`; resultDiv.style.background = 'linear-gradient(135deg, #667eea 0%, #764ba2 100%)'; } // Allow
Enter key to calculate document.querySelectorAll('input[type="number"]').forEach(input => {
input.addEventListener('keypress', function(event) { if (event.key === 'Enter') { calculate(); } }); });
</script> </body> </html>
```

■■ Output Screenshot

Simple
Calculator
JavaScript Event-Driven Programming
Number 1
+
Number 2
Calculate
=?

Question 4

■ Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>Color Changer - JavaScript Events</title> <style>
* { margin: 0; padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma, Geneva,
Verdana, sans-serif; background: linear-gradient(135deg, #2C3E50 0%, #4CA1AF 100%); min-height: 100vh;
display: flex; align-items: center; justify-content: center; padding: 20px; } .container { background:
white; padding: 60px 80px; border-radius: 25px; box-shadow: 0 20px 60px rgba(0,0,0,0.3); text-align:
center; max-width: 600px; width: 100%; } h1 { font-size: 2.5em; color: #2c3e50; margin-bottom: 15px; }
.subtitle { color: #7f8c8d; font-size: 1.1em; margin-bottom: 40px; } .color-box { width: 100%; height:
300px; background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); border-radius: 20px;
```

```
display: flex; align-items: center; justify-content: center; font-size: 2em; color: white; font-weight:
bold; cursor: pointer; transition: all 0.4s ease; box-shadow: 0 10px 30px rgba(0,0,0,0.2); margin-bottom:
30px; } .color-box:hover { transform: scale(1.05) rotate(2deg); box-shadow: 0 20px 50px rgba(0,0,0,0.3); }
.color-box:active { transform: scale(0.98); } .button-group { display: flex; gap: 15px; flex-wrap: wrap; }
button { flex: 1; min-width: 150px; background: linear-gradient(135deg, #2C3E50 0%, #4CA1AF 100%); color:
white; border: none; padding: 18px 30px; font-size: 1.2em; border-radius: 12px; cursor: pointer;
transition: all 0.3s ease; font-weight: 600; } button:hover { transform: translateY(-3px); box-shadow: 0
10px 30px rgba(44, 62, 80, 0.4); } button:active { transform: translateY(0); } #colorCode { margin-top:
20px;
```

```
padding: 15px; background: #f0f2f5; border-radius: 10px; font-size: 1.3em; font-weight: bold; color:
#2c3e50; } @media (max-width: 600px) { .container { padding: 40px 30px; } .color-box { height: 250px;
font-size: 1.5em; } button { width: 100%; } } </style> </head> <body> <div class="container"> <h1>■ Color
Changer</h1> <p class="subtitle">JavaScript Event-Driven Programming</p> <div class="color-box"
id="colorBox" onmouseover="showHoverEffect()" onmouseout="hideHoverEffect()" onclick="changeColor()">
Hover & Click Me! ■■ </div> <div class="button-group"> <button onclick="changeColor()">■ Random
Color</button> <button onclick="resetColor()">■ Reset</button> </div> <div id="colorCode">#667eea →
#764ba2</div> </div> <script> const defaultGradient = 'linear-gradient(135deg, #667eea 0%, #764ba2 100%)';
let currentColor = '#667eea';
```

```
function getRandomColor() { return '#' + Math.floor(Math.random() * 16777215).toString(16).padStart(6,
'0'); } function changeColor() { const color1 = getRandomColor(); const color2 = getRandomColor(); const
gradient = `linear-gradient(135deg, ${color1} 0%, ${color2} 100%)`; const box =
document.getElementById('colorBox'); box.style.background = gradient; currentColor = color1;
document.getElementById('colorCode').textContent = `${color1} → ${color2}`; } function resetColor() {
const box = document.getElementById('colorBox'); box.style.background = defaultGradient;
document.getElementById('colorCode').textContent = '#667eea → #764ba2'; } function showHoverEffect() {
const box = document.getElementById('colorBox'); box.textContent = '■ Click for Random Colors!'; }
function hideHoverEffect() { const box = document.getElementById('colorBox'); box.textContent = 'Hover &
Click Me! ■■'; } </script> </body> </html>
```

■■ Output Screenshot

Color Changer
JavaScript Event-Driven Programming
Hover & Click Me!
Random Color
Reset
#667eea → #764ba2

Question 5

■ Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>Event Log - JavaScript Events</title> <style> * {
margin: 0; padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif; background: linear-gradient(135deg, #8E2DE2 0%, #4A00E0 100%); min-height: 100vh; display:
flex; align-items: center; justify-content: center; padding: 20px; } .container { background: white;
padding: 60px 80px; border-radius: 25px; box-shadow: 0 20px 60px rgba(0,0,0,0.3); text-align: center;
max-width: 700px; width: 100%; } h1 { font-size: 2.5em; color: #2c3e50; margin-bottom: 15px; } .subtitle {
color: #7f8c8d; font-size: 1.1em; margin-bottom: 40px; } input { width: 100%; padding: 20px; font-size:
1.3em; border: 3px solid #8E2DE2;
```

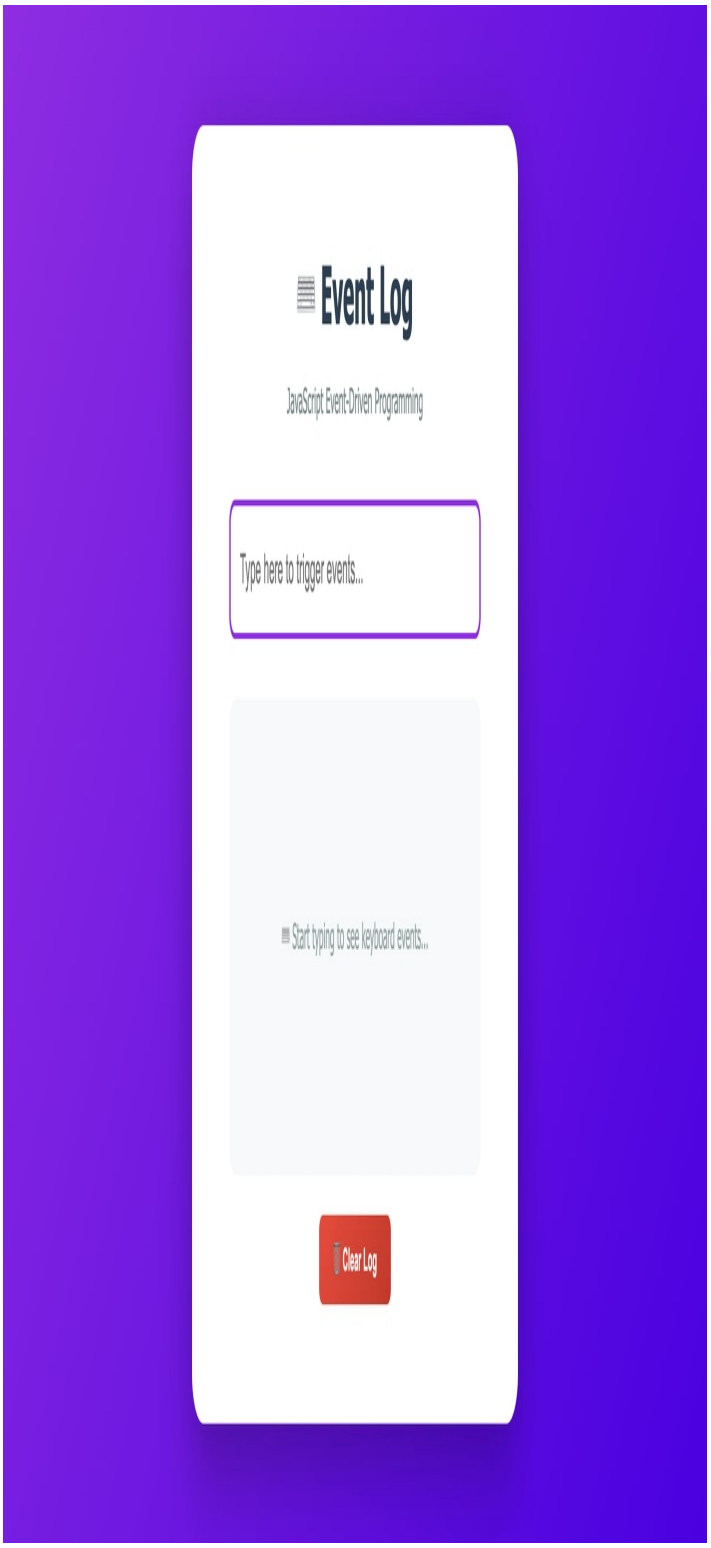
```
border-radius: 12px; margin-bottom: 30px; transition: all 0.3s ease; } input:focus { outline: none;
border-color: #4A00E0; box-shadow: 0 0 20px rgba(142, 45, 226, 0.3); } #output { background: #f8f9fa;
border-radius: 15px; padding: 20px; max-height: 400px; overflow-y: auto; text-align: left; } .event-item {
background: white; padding: 15px 20px; margin: 10px 0; border-left: 5px solid #8E2DE2; border-radius: 8px;
box-shadow: 0 2px 8px rgba(0,0,0,0.08); animation: slideIn 0.3s ease; } @keyframes slideIn { from {
opacity: 0; transform: translateX(-20px); } to { opacity: 1; transform: translateX(0); } } .event-time {
color: #8E2DE2; font-weight: bold; font-size: 0.9em; margin-right: 10px; } .event-key { display:
inline-block; background: linear-gradient(135deg, #8E2DE2 0%, #4A00E0 100%);
```

```
color: white; padding: 5px 12px; border-radius: 6px; font-weight: bold; margin: 0 5px; } .event-type {
color: #7f8c8d; font-size: 0.85em; margin-top: 5px; } .clear-btn { background: linear-gradient(135deg,
#e74c3c 0%, #c0392b 100%); color: white; border: none; padding: 12px 30px; font-size: 1em; border-radius:
8px; cursor: pointer; transition: all 0.3s ease; font-weight: 600; margin-top: 20px; } .clear-btn:hover {
transform: translateY(-2px); box-shadow: 0 8px 20px rgba(231, 76, 60, 0.3); } #output:empty:before {
content: '■■ Start typing to see keyboard events...'; color: #95a5a6; display: flex; align-items: center;
justify-content: center; height: 200px; font-size: 1.1em; } @media (max-width: 600px) { .container {
padding: 40px 30px; } h1 { font-size: 2em; } } </style>
```

```
</head> <body> <div class="container"> <h1>■■ Event Log</h1> <p class="subtitle">JavaScript Event-Driven
Programming</p> <input type="text" id="eventInput" placeholder="Type here to trigger events..."
onkeyup="logEvent(event)" autocomplete="off"> <div id="output"></div> <button class="clear-btn"
onclick="clearLog()">■■ Clear Log</button> </div> <script> let eventCount = 0; function logEvent(event) {
eventCount++; const output = document.getElementById('output'); const time = new
Date().toLocaleTimeString(); const eventItem = document.createElement('div'); eventItem.className =
'event-item'; let keyDisplay = event.key; if (event.key === ' ') keyDisplay = 'Space'; if (event.key ===
'Enter') keyDisplay = 'Enter'; if (event.key === 'Backspace') keyDisplay = 'Backspace';
eventItem.innerHTML = ` <div> <span class="event-time">${time}</span> Event #${eventCount}: Key pressed →
<span class="event-key">${keyDisplay}</span> </div> <div class="event-type"> Type: ${event.type} | Code:
${event.code} | Key: ${event.key} </div> `; output.insertBefore(eventItem, output.firstChild);
```

```
// Limit to 20 events while (output.children.length > 20) { output.removeChild(output.lastChild); } }
function clearLog() { document.getElementById('output').innerHTML = '';
document.getElementById('eventInput').value = ''; eventCount = 0; } </script> </body> </html>
```

■■ Output Screenshot


Event Log
JavaScript Event-Driven Programming
Type here to trigger events...
Start typing to see keyboard events...
Clear Log